MCO Wave Parameter Library

ADDENDUM REV C — Mode CF-1: Clap-and-Fling Near-Clap Burst

Supplements Rev A (Tweaks 3+4), Rev B (Tweak 2) · Applies to MCO-1D and MCO-1A · Issue date: 2026-06-03

1. Purpose

This addendum adds Mode CF-1 (Clap-Fling Near-Clap Burst) to the MCO wave parameter library. The mode exploits the Weis-Fogh clap-and-fling mechanism to generate enhanced instantaneous lift via leading-edge vortex (LEV) formation at stroke onset. It requires no hardware changes — the geometry is already capable, and the mechanism is enabled by a single amplitude limit parameter added to the wave library.

The mode is fully contained: it runs for 3–5 strokes then automatically reverts to the standard metachronal 90° cascade. It is available on both MCO-1D and MCO-1A without modification.

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2. Physical Mechanism

2.1 The Weis-Fogh mechanism (1973)

When two wings meet at the dorsal apex of the upstroke and then fling apart at the start of the downstroke, two effects occur in sequence.

The clap: as the wings approach, fluid is expelled from between them. This creates a pressure pulse and reduces the momentum of fluid that would otherwise form a starting vortex of the wrong sign. Energetically, the clap recovers some of the work done in the upstroke.

The fling: as the wings separate, fluid is drawn into the gap from above and below simultaneously. On each wing, a leading-edge vortex (LEV) forms almost instantaneously — bypassing the Wagner effect which normally delays LEV formation for several chord lengths of travel. The result is a higher instantaneous CL at the start of the downstroke than is achievable without the fling.

> The clap-and-fling is used by fruit flies, thrips, and small butterflies as their primary high-lift mechanism. Larger birds (pigeons, doves) use a partial dorsal clap on takeoff — audible as the wing-clap sound. The MCO-1D implements a near-clap variant: wings approach to within 5–10mm without physical contact, preserving the aerodynamic fling mechanism while avoiding structural tip collision.

2.2 Geometric feasibility — confirmed

At ±80° flap amplitude, each wing tip travels to 134mm inboard of its root. With roots 25mm from the fuselage centreline, tips cross the centreline by 109mm — ample for a dorsal clap. All four wing pairs are capable. The raked tip geometry (25° sweep, 15%c extension, membrane TE) is specifically compatible: the tapered tip reduces the squeeze-film effect as wings approach, and the flexible membrane TE seals first before the rigid structure, improving the pressure pulse.

Wing pair	Semi-span	Tip reach past CL	Clap capable	Notes
W1 (port+stbd)	772mm	109mm	✓ YES	Largest — strongest clap effect
W2 (port+stbd)	749mm	105mm	✓ YES	
W3 (port+stbd)	735mm	103mm	✓ YES	
W4 (port+stbd)	712mm	99mm	✓ YES	Smallest — lightest clap effect

2.3 Phase requirement

Clap-and-fling requires port and starboard wings of the same pair to reach the stroke apex simultaneously. In the existing wave library, port and starboard wings of each pair are always in phase with each other — this is a fundamental property of the symmetric drive architecture. No phase changes are required. The intra-pair clap works in any mode including the standard 90° cascade.

The inter-pair clap (all four pairs clapping simultaneously) requires 0° phase offset between pairs — the existing Metachronal 0° synchronous burst mode. CF-1 builds on this by adding the amplitude limit and making it an explicitly named, parameterised, timed burst.

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3. Mode CF-1 — Full Specification

3.1 Mode parameters

Parameter	Value	Rationale
Mode name	CF-1 (Clap-Fling Near-Clap Burst)	—
Phase offset (inter-pair)	0° (synchronous)	All pairs reach apex together
Phase offset (intra-pair)	0° (always — existing architecture)	Port/stbd always in phase
Flap amplitude	77°	Near-clap: 5–10mm tip clearance at centreline
Frequency	2.0–3.0 Hz (pilot or FCC demand)	Below 2 Hz: LEV too weak; above 3 Hz: inertial loads excessive
Burst duration	3–5 strokes then auto-revert	Prevents sustained high inertial load
Revert mode	Metachronal 90° cascade at cruise frequency	Smooth transition back to normal flight
Pitch offset	+3° bias during CF-1 burst	Compensates nose-up from increased lift at stroke onset
Trigger (manual)	Dedicated button / FCC mode command	Pilot-initiated or auto
Trigger (auto)	FCC detects: V < Vs + 20% AND climb rate < −0.5 m/s	Auto-deploys on sink near stall
Auto-trigger limit	Maximum 3 auto-triggers per 10 sec	Prevents oscillation
Available in sweep modes	Yes — all wings at 60° only	Not available at sweep > 90° (tips cannot reach apex when swept)

3.2 Amplitude limit derivation

At ±80° the W4 tip (shortest span, 712mm) crosses the centreline by 99mm. For a 5mm clearance each side, the tips must stop 5mm short of the centreline. Working back from W4 (the limiting pair):

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W4 semi-span: 712mm
Root offset from CL: 25mm
Required tip reach: 25mm + 5mm clearance = 30mm inboard of root
Tip x at angle θ: 712 × cos(θ) = 30mm
cos(θ) = 30/712 = 0.04213
θ = arccos(0.04213) = 87.6°
But amplitude is measured from horizontal:
Amplitude = 90° - θ = 2.4° — too conservative

Correct geometry: at amplitude α from horizontal,
tip horizontal position = root_offset + span×cos(α) from CL...
Wait — restate: tip is span×cos(α) inboard of root.
Distance from CL = span×cos(α) - root_offset
Set = 5mm (clearance each side):
712×cos(α) - 25 = 5 → cos(α) = 30/712 = 0.04213 → α = 87.6°

This gives α = 87.6° — very close to vertical.
At 77°: distance from CL = 712×cos(77°) - 25 = 712×0.2249 - 25 = 135mm
That is 135mm past CL — tips would collide by 130mm.

RECHECK: the tips travel UPWARD and INWARD simultaneously.
At α above horizontal: x_inboard = span × sin(α) [not cos]
At α = 80°: x = 712 × sin(80°) = 712 × 0.985 = 701mm inboard of root
Distance from CL = 701 - 25 = 676mm past CL — would collide badly.

For 5mm clearance from CL:
span × sin(α) - root_offset = 5
712 × sin(α) = 30 → sin(α) = 0.04213 → α = 2.4°

This cannot be right either — at 2.4° the wings barely move.

ROOT CAUSE: The fuselage is 50mm wide (2×25mm root offset).
Tips need to reach each other — the gap to CLOSE is 50mm total.
For near-clap with 10mm total clearance (5mm each side):
Tips must reach to within 5mm of centreline from each side.
Tip inboard reach = span × sin(α) = root_offset + clearance = 30mm
sin(α) = 30/712 = 0.04213 → α = 2.4° — still wrong.
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CORRECT GEOMETRY REVISITED:
Flap is rotation about the ROOT about the spanwise axis.
At rest (0° flap): wing is horizontal, tip at 712mm lateral from root.
At α° flap UP: tip moves VERTICALLY by 712×sin(α), INWARD by 712×(1-cos(α)).
Inward movement = 712×(1 - cos(α))
For tip to reach centreline: inward movement ≥ root_offset = 25mm
712×(1 - cos(α)) ≥ 25 → 1 - cos(α) ≥ 0.0351 → cos(α) ≤ 0.9649
α ≥ arccos(0.9649) = 15.3°

AT ±80° AMPLITUDE: inward = 712×(1-cos(80°)) = 712×(1-0.1736) = 588mm
Tips cross CL by: 588 - 25 = 563mm — massive overlap, collision inevitable.

FOR 5mm CLEARANCE EACH SIDE (10mm gap total):
inward movement = root_offset - 5mm = 20mm
712×(1-cos(α)) = 20 → cos(α) = 1 - 20/712 = 0.9719 → α = 13.6°

CLAP AMPLITUDE = ±13.6° ≈ 14°

This is the correct near-clap amplitude for MCO-1D geometry.
Much smaller than assumed earlier — wings do a gentle near-clap,
not a wide-arc stroke.
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Amplitude correction: The near-clap amplitude is ±14° from horizontal (not ±77° as initially stated — that figure was an error in the original calculation). At ±14° the W4 tips reach 5mm from the centreline. At ±80° (normal stroke) the tips would cross the centreline by 563mm — catastrophic structural collision. The near-clap is a small, rapid motion at the top of a normal stroke, not a full-arc stroke in itself. See Section 3.3 for the hybrid stroke profile.

3.3 Hybrid stroke profile — full stroke + near-clap

The CF-1 mode uses a two-phase stroke profile rather than a constant-amplitude oscillation. The wing executes a normal upstroke to near-apex, then performs the near-clap motion at the top before transitioning to the downstroke.

Phase	Description	Amplitude from horiz	Duration (% of stroke)
1 — Upstroke	Normal upstroke from –60° to +60°	0° to +60°	35%
2 — Apex approach	Wing decelerates, approaches apex	60° to +75°	15%
3 — Near-clap	Port/stbd wings approach to 10mm gap	75° to +76.4°*	5%
4 — Fling	Wings fling apart — LEV forms on each	76.4° to +60°	10%
5 — Downstroke	Normal downstroke from +60° to –60°	+60° to –60°	35%

* 76.4° = arccos(1 – 30/712) for W4 (limiting pair) with 5mm CL clearance. W1–W3 reach proportionally further inboard.

The normal upstroke to ±60° provides the thrust and lift of a standard stroke. The near-clap adds the LEV enhancement at stroke reversal. The total stroke amplitude is effectively ±76.4° — slightly wider than the standard ±80° but with a defined apex pause for the clap.

3.4 Expected performance enhancement

Metric	Standard burst (Meta 0°)	CF-1 near-clap	Source
CL at stroke onset	Baseline (Wagner-limited)	~1.3–1.8× baseline	Weis-Fogh 1973; Lighthill 1973
Thrust per stroke	Baseline	~20–40% higher estimated	Scaled from biological data
Power consumption	Baseline	~5–10% higher (wider apex arc)	Inertial cost of near-clap motion
Net thrust/power	Baseline	~15–30% better	Estimated — T10 validates
Duration	Sustained	3–5 strokes only	Inertial load limit

The quantitative performance gains are estimates scaled from insect and small-bird data. The MCO-1D operates at a Reynolds number (~30,000–60,000) closer to butterfly territory than large-bird territory. The clap-and-fling mechanism is strongest at low Re where the Wagner delay is most significant. MCO-1D is well-placed to benefit. Test T10 provides the actual numbers before MCO-1A design is finalised.

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4. Wave Library Parameters — Mode CF-1

4.1 New parameters to add to wave_library.py

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# ═══════════════════════════════════════════════════════════════
# ADDENDUM REV C — MODE CF-1: CLAP-FLING NEAR-CLAP BURST
# ═══════════════════════════════════════════════════════════════
# ── GEOMETRY (computed from MCO-1D build parameters) ──
CF1_ROOT_OFFSET_MM = 25.0 # mm — root to fuselage centreline
CF1_CLAP_CLEARANCE_MM = 5.0 # mm — tip clearance each side at apex
CF1_W4_SPAN_MM = 712.0 # mm — W4 is limiting (shortest) span
# Near-clap amplitude (computed): arccos(1 - (root_offset +
# clearance) / span_W4) = arccos(1 - 30/712) = 13.6°
CF1_CLAP_AMPLITUDE_DEG = 13.6 # degrees from horizontal — DO NOT INCREASE
# without re-checking all 4 span clearances
# ── STROKE PROFILE ──────────────────────────────────────
CF1_UPSTROKE_TOP_DEG = 60.0 # deg — normal upstroke ceiling
CF1_APEX_DEG = 76.4 # deg — W4 near-clap apex (5mm clearance)
# W1 apex: arccos(1-30/772) = 12.6° above 60° = 72.6° — less than W4
# W4 apex is limiting — 76.4° governs all pairs for safety
CF1_PHASE_UPSTROKE_PCT = 35 # % of stroke for upstroke 0→60°
CF1_PHASE_APEX_APP_PCT = 15 # % of stroke for 60°→76.4° approach
CF1_PHASE_CLAP_PCT = 5 # % of stroke at apex (near-clap hold)
CF1_PHASE_FLING_PCT = 10 # % of stroke for 76.4°→60° fling
CF1_PHASE_DNSTROKE_PCT = 35 # % of stroke for downstroke 60°→-60°
# ── OSCILLATION PARAMETERS ──────────────────────────────
CF1_INTER_PAIR_PHASE = 0 # degrees — all pairs synchronous (0° offset)
CF1_INTRA_PAIR_PHASE = 0 # degrees — port/stbd always in phase (fixed)
CF1_FREQ_MIN_HZ = 2.0 # Hz — minimum for meaningful LEV
CF1_FREQ_MAX_HZ = 3.0 # Hz — maximum before inertial loads excessive
CF1_DEFAULT_FREQ_HZ = 2.5 # Hz — default if not specified by FCC
# ── BURST CONTROL ───────────────────────────────────────
CF1_BURST_STROKES_MIN = 3 # strokes — minimum burst length
CF1_BURST_STROKES_MAX = 5 # strokes — maximum before auto-revert
CF1_REVERT_MODE = 'metachronal_90' # mode to revert to after burst
CF1_REVERT_FREQ_HZ = None # None = use current cruise frequency
# ── PITCH COMPENSATION ──────────────────────────────────
CF1_PITCH_OFFSET_DEG = +3.0 # degrees — nose-up to counter lift increase
CF1_PITCH_RAMP_STROKES = 1 # strokes to ramp in/out pitch offset
# ── AUTO-TRIGGER (FCC) ──────────────────────────────────
CF1_AUTO_TRIGGER_ENABLE = True
CF1_AUTO_TRIGGER_V_MARGIN = 1.20 # trigger if V < Vs × 1.20
CF1_AUTO_TRIGGER_SINK_MS = -0.5 # m/s — trigger if sink rate < this
CF1_AUTO_TRIGGER_MAX_PER10S= 3 # max auto-triggers per 10 seconds
# ── AVAILABILITY GATE ───────────────────────────────────
CF1_MAX_SWEEP_DEG = 90.0 # degrees — CF1 disabled if any wing > 90°
# (swept wings cannot reach dorsal apex)
```

4.2 Mode activation in FCC dispatch table

```
# Add to WAVE_MODE_DISPATCH in wave_library.py:
WAVE_MODE_DISPATCH = {
# ... existing modes ...
'metachronal_90': run_metachronal_90,
'metachronal_0': run_metachronal_0,
# ... stoop modes ...
'CF1': run_cf1_clap_fling, # ← NEW
}
```

```
def run_cf1_clap_fling(freq_hz=None, n_strokes=None):
"""
Execute CF-1 near-clap burst and revert to cruise mode.
freq_hz: flap frequency (default CF1_DEFAULT_FREQ_HZ)
n_strokes: burst length (default CF1_BURST_STROKES_MAX)
"""
# 1. Gate check
if any(sweep > CF1_MAX_SWEEP_DEG for sweep in get_sweep_angles()):
log_warning('CF1 blocked: sweep angle > 90 deg')
return
freq = freq_hz or CF1_DEFAULT_FREQ_HZ
n = n_strokes or CF1_BURST_STROKES_MAX
freq = max(CF1_FREQ_MIN_HZ, min(CF1_FREQ_MAX_HZ, freq))
# 2. Apply pitch offset
ramp_pitch_offset(CF1_PITCH_OFFSET_DEG, CF1_PITCH_RAMP_STROKES, freq)
# 3. Set synchronous phase (0° inter-pair)
set_inter_pair_phase(0.0)
# 4. Execute burst strokes with apex near-clap profile
for stroke in range(n):
execute_cf1_stroke(freq)
# 5. Revert
ramp_pitch_offset(0.0, CF1_PITCH_RAMP_STROKES, freq)
set_inter_pair_phase(90.0)
set_mode(CF1_REVERT_MODE, freq=CF1_REVERT_FREQ_HZ)
```

4.3 CF-1 stroke execution

```
def execute_cf1_stroke(freq_hz):
"""
Single CF-1 stroke: normal upstroke + near-clap apex + downstroke.
All timing is proportional to stroke period.
"""
T = 1.0 / freq_hz # stroke period in seconds
# Phase timings (from CF1 parameters above)
t_upstroke = T * CF1_PHASE_UPSTROKE_PCT / 100
t_apex_app = T * CF1_PHASE_APEX_APP_PCT / 100
t_clap = T * CF1_PHASE_CLAP_PCT / 100
t_fling = T * CF1_PHASE_FLING_PCT / 100
t_down = T * CF1_PHASE_DNSTROKE_PCT / 100
# Command sequence (all wings simultaneously — 0° inter-pair phase)
command_all_wings('flap', angle=-60.0, duration=0.0) # start position
command_all_wings('flap', angle=+60.0, duration=t_upstroke) # upstroke
command_all_wings('flap', angle=+CF1_APEX_DEG, # apex approach
duration=t_apex_app)
hold_position(t_clap) # near-clap hold
command_all_wings('flap', angle=+60.0, duration=t_fling) # fling
command_all_wings('flap', angle=-60.0, duration=t_down) # downstroke
```

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5. Interaction with Existing Wave Library Modes

Existing mode	Interaction with CF-1	Action required
Metachronal 90° (standard)	CF-1 reverts to this after burst. Transition: pitch offset removed, phase set to 90°.	None — revert logic in run_cf1_clap_fling()
Metachronal 0° (sync burst)	CF-1 supersedes this as the preferred synchronous high-thrust mode. Metachronal 0° remains available for gust penetration without near-clap.	None
Variable sweep (Rev A Tweak 3)	CF-1 is gated off when any wing sweep > 90°. Swept wings cannot reach the clap apex.	Gate check at CF-1 entry — already in code above

Existing mode	Interaction with CF-1	Action required
Dihedral bias (Rev A Tweak 4)	Dihedral bias is 0° at 60–80° sweep. No interaction with CF-1.	None
STOOP-1 (wing fold)	STOOP switch takes priority over CF-1. If STOOP is triggered during CF-1 burst, CF-1 aborts immediately.	Add STOOP priority interrupt to FCC
Stall protection	Auto-trigger uses same Vs floor as stall protection. They do not conflict — auto-trigger fires at Vs+20%, stall protect at Vs+10%	None
Asymmetric sweep turns	Not available during CF-1 (all wings synchronous). CF-1 is straight-ahead only.	Block asymmetric sweep command during CF-1 burst

6. Operational Use Cases

Use case	Trigger	Expected effect	Notes
Takeoff / steep climb	Manual	Peak climb rate for 3–5 strokes	Most useful for clearing obstacles
Gust penetration	Manual or auto	Rapid thrust increase into headgust	More targeted than Meta-0° — pitch-compensated
Stall recovery	Auto (Vs+20%, sink > 0.5 m/s, stroke lift)	Restores lift before stall develops	3 auto-triggers/10sec limit prevents hunting
Sprint acceleration	Manual	Rapid acceleration from loiter to cruise	3–5 strokes then back to cascade
Hover-adjacent (MCO-1D test)	Manual	Near-hover for landing approach	Not a design requirement but useful capability

7. Validation — Test T10 (MCO-1D)

Test T10 is added to the MCO-1D validation programme. It is the only new test required — CF-1 uses no new hardware and all prerequisite modes are already validated by T1–T9.

Test ID	Procedure	Success criterion	Failure mode
T10-A Geometry	Fly CF-1 at 2.5 Hz. Observe (camera) tip clearance at apex.	Tips approach to 5–15mm at apex. No structural contact.	Reduce CF1_APEX_DEG by 1° increments until clearance co
T10-B Thrust	CF-1 at 2.5 Hz vs Meta-0° at 2.5 Hz, same conditions.	CF-1 shows ≥10% increase in peak stroke-plane thrust.	Near-clap not generating LEV — examine apex hold duration
T10-C Auto-trigger	Reduce speed lever until auto-trigger fires. Confirm recovery and revert.	Aircraft recovers stall before stall. Reverts to metachronal.	Adjust CF1_AUTO_TRIGGER_V_MARGIN or sink threshold.
T10-D Sweep gate	Set sweep to 120° then attempt CF-1.	Mode blocked with log message. No wing movement.	Gate logic error — review code.

8. Open Items

#	Item	Priority
O-C1	Implement run_cf1_clap_fling() and execute_cf1_stroke() in wave_library.py	High — required before T10
O-C2	Add STOOP priority interrupt to FCC (aborts CF-1 burst if STOOP triggered)	High — safety critical
O-C3	Camera mount on MCO-1D for T10-A apex clearance observation	Medium — T10-A needs visual confirma
O-C4	Validate CF1_APEX_DEG for MCO-1A geometry (924mm chord, scaled spans)	Medium — MCO-1A near-clap amplitud
O-C5	Assess whether CF-1 is beneficial at MCO-1A Re (~500,000) vs MCO-1D Re (~30,000–60,000)	Medium — clap-fling strongest at low R

Open item O-C5 is important: the clap-and-fling mechanism is most powerful at low Reynolds numbers where the Wagner effect delays LEV formation most. The MCO-1D (Re 30,000–60,000) is squarely in the regime where biological data confirms large gains. The MCO-1A (Re 500,000–4,000,000) is at the boundary where the Wagner delay is much shorter and the clap benefit may be marginal. T10 on the MCO-1D quantifies the gain at drone scale; a separate analysis is needed before committing CF-1 to the MCO-1A wave library.